

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

End-to-End Data Analytics & Business Intelligence Project

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Tools Used

Excel | Python | SQL | Power BI | DAX

Year

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1. PROJECT OVERVIEW

Customer Shopping Behavior Analysis is an end-to-end Data Analytics and Business Intelligence project developed to understand customer purchasing behavior, revenue performance, product performance, and customer characteristics.

The project uses a customer shopping dataset containing **3,900 records and 18 attributes**. The dataset includes customer demographics, products purchased, purchase amounts, locations, seasons, review ratings, subscription status, shipping types, discounts, payment methods, previous purchases, and purchase frequency.

The project was completed through a structured analytics workflow using:

Excel → Python → SQL → Power BI

Excel was used for the initial inspection and understanding of the dataset. Python was used for data preparation and exploratory analysis. SQL was used to query the prepared data and perform business-oriented analysis. Power BI was then used to create analytical measures, KPIs, interactive visualizations, slicers, and dashboards.

The final Power BI solution consists of four pages:

1. Executive Overview
2. Customer Analysis
3. Product & Sales Analysis
4. Final Executive Dashboard

The project focuses on turning raw customer shopping data into useful information that can help understand sales performance, customer behavior, and product-related patterns.

2. PROBLEM STATEMENT

Businesses collect large amounts of customer shopping data, but raw data by itself does not provide clear information for decision-making.

It can be difficult to understand:

- Which product categories generate the most revenue
- Which products perform strongly
- Which customer groups contribute more to revenue
- How customers differ by age and gender
- How purchasing frequency relates to customer behavior
- How customer segments behave
- How revenue differs across seasons
- Which payment and shipping methods are commonly used
- How discounts relate to purchases
- Which locations, sizes, and colors perform better

The purpose of this project is to analyze the available customer shopping data and develop an interactive Business Intelligence dashboard that allows users to explore these patterns and support data-driven decision-making.

3. PROJECT OBJECTIVES

The main objectives of the project are:

1. Understand the structure and characteristics of the customer shopping dataset.
2. Inspect and prepare the data before analysis.
3. Clean and validate the dataset using appropriate data analytics tools.
4. Perform exploratory analysis to understand customer and purchasing patterns.
5. Use SQL to query the data and answer business-oriented questions.
6. Analyze overall revenue and purchase performance.
7. Understand customer demographics and purchasing behavior.
8. Analyze customer purchasing frequency.
9. Analyze customer segments and subscription behavior.
10. Identify high-performing product categories and products.
11. Analyze revenue by gender and age group.
12. Compare revenue across seasons and locations.
13. Analyze payment methods and shipping types.
14. Analyze the use of discounts.
15. Analyze product attributes such as size and color.
16. Create meaningful KPIs using DAX.
17. Build interactive Power BI dashboards.
18. Present the most important findings through an executive dashboard.
19. Generate business recommendations based on the analysis.

4. PROJECT WORKFLOW

The project followed an end-to-end Data Analyst workflow.

Overall Process

Raw Dataset

↓

Excel — Initial Inspection

↓

Python — Cleaning & Exploratory Analysis

↓

SQL — Business Analysis & Queries

↓

Power BI / Power Query — Data Preparation

↓

DAX — KPI & Measure Creation

↓

Power BI — Dashboard Development

↓

Interactive Dashboard

↓

Business Insights & Recommendations

4.1 Excel

Excel was used as the starting point for understanding the dataset.

The dataset was inspected to understand:

- Number of records
- Number of columns
- Column names
- Numerical fields
- Categorical fields
- Possible missing values
- Overall data structure

Excel provided an initial view of the raw data before moving into programmatic analysis.

4.2 Python

Python was used for data preparation and exploratory data analysis.

The Python stage focused on:

- Loading the dataset
- Inspecting the dataset structure
- Checking rows and columns
- Reviewing data types
- Checking missing values
- Examining numerical columns

- Reviewing categorical variables
- Exploring customer and purchasing patterns
- Supporting the analysis before visualization

Python was useful for understanding the dataset programmatically and identifying patterns that could later be explored using SQL and Power BI.

4.3 SQL

SQL was used to query the prepared data and perform business-oriented analysis.

The SQL analysis focused on questions such as:

- What is the overall revenue?
- How many purchases are present?
- What is the average purchase amount?
- Which categories generate higher revenue?
- How does revenue vary by gender?
- How does revenue vary by age group?
- Which customer segments contribute more revenue?
- How does purchasing frequency affect revenue?
- How does revenue vary by season?
- Which payment methods are associated with purchases?
- How does discount usage compare?
- Which locations generate higher revenue?
- Which sizes and colors perform better?
- Which products generate higher revenue?

SQL provided a structured way to aggregate, filter, group, and compare the data before creating the final dashboard.

5. DATASET DESCRIPTION

5.1 Dataset Overview

The dataset contains:

- **3,900 records**
- **18 attributes**

The dataset represents customer shopping activity and includes information related to:

- Customers

- Products
- Purchases
- Demographics
- Customer behavior
- Payment
- Shipping
- Discounts

The dataset does **not contain a transaction date or purchase date**. Therefore, the project does not include a true monthly/weekly revenue trend analysis.

5.2 Dataset Fields

Field	Description
Customer ID	Unique identifier for the customer
Age	Age of the customer
Gender	Gender of the customer
Item Purchased	Product purchased
Category	Product category
Purchase Amount (USD)	Amount spent on the purchase
Location	Customer location
Size	Size of the purchased product
Color	Color of the purchased product
Season	Season associated with the purchase
Review Rating	Customer review rating
Subscription Status	Indicates whether the customer has a subscription
Shipping Type	Shipping method selected
Discount Applied	Indicates whether a discount was applied
Promo Code Used	Indicates whether a promotional code was used
Previous Purchases	Number of previous purchases
Payment Method	Payment method used
Frequency of Purchases	Frequency at which purchases are made

6. DATA UNDERSTANDING AND INITIAL INSPECTION

Before performing detailed analysis, the dataset was inspected to understand its structure and the type of information available.

The initial inspection focused on:

- Number of records
- Number of columns
- Column names
- Data types
- Numerical fields
- Categorical fields
- Missing or blank values
- Unusual values
- Customer-related attributes
- Product-related attributes
- Purchase-related attributes

The dataset was confirmed to contain **3,900 records and 18 attributes**.

7. DATA CLEANING AND PREPARATION

Data preparation was performed before the dataset was used for business analysis and visualization.

7.1 Excel Data Inspection

Excel was used for the initial inspection of the raw dataset.

The following were reviewed:

- Dataset structure
- Column names
- Number of records
- Data types
- Numerical fields
- Categorical fields
- Missing or blank values
- General data consistency

This helped establish an understanding of the data before moving to Python.

7.2 Python Data Preparation

Python was used for programmatic data inspection and preparation.

The process included:

- Loading the dataset into Python
- Checking the shape of the dataset
- Reviewing column names
- Inspecting data types
- Checking missing values
- Reviewing numerical columns
- Examining categorical columns
- Checking the data before analysis

Python also supported exploratory analysis of the available customer, product, and purchasing variables.

7.3 Data Quality Checks

The following areas were reviewed during data preparation:

- Column names and structure
- Numerical data fields
- Categorical data fields
- Review ratings
- Missing or blank values
- Data types
- Categorical consistency
- Purchase-related fields

The **Review Rating** field was specifically reviewed for blank or missing values during data inspection.

8. EXPLORATORY DATA ANALYSIS USING PYTHON

Exploratory Data Analysis was used to understand the main characteristics of the dataset before creating the final dashboard.

The analysis focused on:

Customer Analysis

- Age distribution

- Gender distribution
- Customer purchasing behavior
- Previous purchases
- Purchase frequency
- Subscription status

Product Analysis

- Product categories
- Items purchased
- Product sizes
- Product colors
- Product performance

Revenue Analysis

- Total purchase amount
- Average purchase amount
- Revenue by category
- Revenue by gender
- Revenue by customer characteristics
- Revenue by season
- Revenue by location

Purchase Behavior

- Frequency of purchases
- Discounts
- Promo code usage
- Payment methods
- Shipping types

The exploratory stage helped identify the dimensions that were most useful for the later SQL analysis and Power BI dashboard.

9. SQL ANALYSIS

SQL was used to perform structured analysis on the prepared dataset.

The analysis used aggregation, grouping, filtering, and comparison to answer business questions.

9.1 Revenue Analysis

Revenue was analyzed to understand overall sales performance and compare revenue across different dimensions.

Key areas included:

- Total revenue
- Average purchase amount
- Revenue by category
- Revenue by gender
- Revenue by age group
- Revenue by customer segment
- Revenue by season

9.2 Customer Analysis

Customer-related SQL analysis included:

- Customer count
- Age-related analysis
- Gender analysis
- Purchase frequency
- Previous purchases
- Subscription status
- Customer segments

9.3 Product Analysis

Product analysis focused on:

- Items purchased
- Product categories
- Top products
- Product sizes
- Product colors

9.4 Sales and Purchase Behavior

The analysis also examined:

- Payment methods
- Shipping types

- Discount usage
- Promo code usage
- Locations
- Seasonal behavior

9.5 SQL Analysis Dimensions

The project used the following major analytical dimensions:

- Category
- Gender
- Age / Age Group
- Customer Segment
- Frequency of Purchases
- Season
- Payment Method
- Shipping Type
- Discount Applied
- Location
- Size
- Color
- Item Purchased

These dimensions were later used in Power BI to create comparative visualizations.

10. POWER BI DATA PREPARATION

After the Excel, Python, and SQL stages, the prepared dataset was brought into Power BI.

Power Query was used within the Power BI workflow to inspect and prepare the data for visualization.

The process included:

- Loading the dataset
- Reviewing data types
- Checking fields required for analysis
- Ensuring numerical fields could be used for calculations
- Ensuring categorical fields could be used for grouping and filtering
- Preparing the dataset for Power BI analysis

The prepared data was then used to build the dashboard.

11. DATA MODELING

The project primarily uses a single customer shopping dataset containing customer, product, purchase, and behavioral information.

Because the required analytical attributes were available within the same dataset, a complex multi-table relational model was not required.

The dataset was loaded into Power BI as the primary analytical table.

The model supports:

- KPI calculations
- Revenue analysis
- Customer analysis
- Product analysis
- Category analysis
- Slicer filtering
- Interactive visualizations

The single-table structure was suitable for this project because the focus was on analytical reporting rather than building a complex transactional data warehouse.

12. DAX MEASURES AND KPIS

DAX was used in Power BI to create six core measures used across the dashboard.

12.1 Average Age

Purpose: Calculates the average age of customers.

Average Age =

AVERAGE('Customer Shopping'[Age])

12.2 Average Purchase

Purpose: Calculates the average amount spent per purchase.

Average Purchase =

AVERAGE('Customer Shopping'[Purchase Amount (USD)])

12.3 Average Rating

Purpose: Calculates the average customer review rating.

Average Rating =

AVERAGE('Customer Shopping'[Review Rating])

12.4 Total Customers

Purpose: Calculates the number of unique customers using Customer ID.

Total Customers =

DISTINCTCOUNT('Customer Shopping'[Customer ID])

12.5 Total Purchases

Purpose: Counts the number of purchase records.

Total Purchases =

COUNTROWS('Customer Shopping')

12.6 Total Revenue

Purpose: Calculates total revenue from purchase amounts.

Total Revenue =

SUM('Customer Shopping'[Purchase Amount (USD)])

12.7 KPI Summary

KPI	Purpose
Average Age	Measures average customer age
Average Purchase	Measures average spending per purchase
Average Rating	Measures average customer rating
Total Customers	Counts unique customers
Total Purchases	Counts purchase records
Total Revenue	Calculates total revenue

The dashboard displays approximately:

- **\$233.08K Total Revenue**
- **3.9K Total Purchases**
- **\$59.76 Average Purchase**

- **3.75 Average Rating**

The customer count is calculated using distinct Customer IDs.

13. DASHBOARD DEVELOPMENT

The Power BI dashboard was developed after completing the data preparation and analytical stages.

The dashboard was designed to provide both detailed analysis and an executive-level summary.

The final solution contains **four pages**.

13.1 PAGE 1 — EXECUTIVE OVERVIEW

Purpose

The Executive Overview provides a high-level view of revenue, purchases, customers, and customer-related patterns.

KPI Cards

The page contains:

1. Total Revenue
2. Total Purchases
3. Average Purchase
4. Average Rating
5. Total Customers

Visualizations

Revenue by Category

Shows how total revenue is distributed across product categories.

Revenue by Age Group

Compares revenue across different customer age groups.

Revenue by Gender

Compares revenue contribution across genders.

Revenue by Customer Segment

Shows revenue generated by different customer segments.

Revenue by Purchase Frequency

Shows revenue across different purchase frequency categories.

Slicers

The page includes:

- Category

- Gender
- Age Group
- Customer Segment

The slicers allow users to filter the dashboard interactively.

Screenshot

Page 1 screenshot:



13.2 PAGE 2 — CUSTOMER ANALYSIS

Purpose

The Customer Analysis page focuses on customer demographics, purchasing behavior, subscription status, and customer segmentation.

KPI Cards

The page contains:

- Total Customers
- Average Age
- Average Purchase
- Average Rating

Visualizations

Customer Count by Age Group

Shows the distribution of customers across different age groups.

Customer Subscription Status

Shows the distribution between subscribed and non-subscribed customers.

Customers by Purchase Frequency

Shows how customers are distributed across different purchasing frequency groups.

Customer Distribution by Gender

Shows customer distribution by gender.

Customers by Segment

Shows the distribution of customers across the identified customer segments.

Average Purchase by Customer Segment

Compares average purchase amounts across customer segments.

This helps identify differences in spending behavior between segments.

Slicers

Interactive slicers allow users to explore customer behavior from different perspectives.

Screenshot

Page 2 screenshot:



13.3 PAGE 3 — PRODUCT & SALES ANALYSIS

Purpose

The Product & Sales Analysis page focuses on product performance and sales-related factors.

KPI Cards

The page contains:

- Total Revenue
- Total Purchases
- Average Purchase
- Average Rating

Visualizations

Top 10 Products by Revenue

Identifies the products generating the highest revenue.

Top 10 Locations by Revenue

Compares revenue across the highest-performing locations.

Revenue by Payment Method

Shows revenue distribution across payment methods.

Revenue by Discount Status

Compares revenue based on discount usage.

Revenue by Season

Compares revenue across seasons.

Revenue by Shipping Type

Shows revenue associated with different shipping methods.

Revenue by Size

Analyzes revenue across product sizes.

Top 10 Colors by Revenue

Identifies the colors associated with the highest revenue.

Screenshot

Page 3 screenshot:



13.4 PAGE 4 — FINAL EXECUTIVE DASHBOARD

Purpose

The Final Executive Dashboard was created as the main presentation-level dashboard.

It provides a concise view of the most important customer, revenue, and product metrics.

The page was designed to resemble a professional executive dashboard while still using the actual data and analysis from the project.

KPI Cards

The page contains:

- Total Revenue
- Total Purchases
- Average Purchase
- Average Rating
- Total Customers

Slicers

The final dashboard contains:

- Gender
- Age Group
- Customer Segment

Visualizations

Revenue by Category

Provides a quick comparison of revenue across product categories.

Revenue by Season

Shows revenue distribution across seasons.

Top 10 Products by Revenue

Highlights the products contributing the highest revenue.

Revenue by Gender

Provides a demographic comparison of revenue.

Revenue by Age Group

Shows revenue contribution across different age groups.

Revenue by Customer Segment

Shows revenue contribution from different customer segments.

Dashboard Design

The final dashboard uses:

- Light blue-gray background
- White header
- White KPI cards
- Dark navy title
- Consistent blue chart colors
- Consistent card sizing
- Consistent slicer sizing
- Aligned visuals
- Data labels
- Number formatting
- Interactive slicers

Screenshot

Page 4 screenshot:



14. DASHBOARD FORMATTING AND INTERACTIVITY

Formatting was performed after the analytical visuals were created.

KPI Formatting

The KPI cards were standardized for:

- Equal width
- Equal height
- Consistent spacing
- Consistent background
- Consistent borders
- Readable values
- Clear KPI labels

Slicer Formatting

Slicers were formatted to maintain:

- Consistent width
- Consistent height
- Consistent appearance
- Easy filtering

The slicers were also tested to ensure that filtering worked correctly.

Chart Formatting

Charts were formatted for:

- Clear titles
- Data labels where appropriate
- Consistent number formatting
- Readable text
- Consistent visual sizing
- Appropriate spacing
- Consistent colors

Large numeric values were formatted using compact representations such as **K** where appropriate.

Final Theme and Background

The final executive dashboard uses a light blue-gray page background with white visual areas and a dark navy title.

This creates separation between the dashboard canvas and individual visual components without making the page overly colorful.

Interactivity

The dashboard uses slicers to allow users to dynamically filter the data.

The final dashboard slicers were tested to confirm that the relevant KPI cards and visualizations respond to selections.

15. KEY BUSINESS INSIGHTS

The final analysis provided several useful insights into customer spending, product performance, and purchasing behavior.

Overall Revenue and Purchase Performance

The dataset contains **3,900 purchase records** and generates approximately **\$233.08K in total revenue**. The average purchase amount is approximately **\$59.76**, while the average customer review rating is **3.75 out of 5**.

Category Performance

Clothing generated the highest revenue at approximately **\$104K**, followed by **Accessories at \$74K**. Footwear generated approximately **\$36K**, while Outerwear contributed approximately **\$19K**.

This shows that Clothing and Accessories were the strongest-performing product categories in terms of revenue.

Customer Demographics

Revenue differed noticeably across genders. **Male customers generated approximately \$158K**, while **female customers generated approximately \$75K**.

Revenue also varied across age groups. The **40–59 age group generated approximately \$90K**, followed by the **20–39 group at approximately \$87K**. The 60–79 group generated approximately \$47K, while customers under 20 generated approximately \$9K.

Customer Segment Performance

The **Occasional** customer segment generated the highest revenue at approximately \$93K, followed by the **Regular** segment at approximately \$87K.

The **Low Engagement** and **High Value** segments generated approximately \$27K and \$26K, respectively.

This indicates that customer segments differ considerably in their contribution to overall revenue.

Purchase Frequency

Revenue was distributed relatively evenly across the purchase-frequency groups. **Every 3 Months** recorded the highest revenue at approximately \$35K, followed by **Annually and Quarterly at approximately \$34K each**. Weekly and Fortnightly purchasing patterns generated approximately \$32K each.

Product Performance

The Top 10 Products analysis highlights the products contributing the highest revenue. In the final dashboard, **Blouse, Shirt, Dress, Pants, and Jewellery** appear among the highest-revenue products, with the displayed values rounded to approximately \$10K each. The visual can be used to compare the revenue contribution of the top-performing products.

Seasonal Performance

Revenue remained relatively consistent across the four seasons.

- **Fall:** approximately \$60K
- **Spring:** approximately \$59K
- **Winter:** approximately \$59K
- **Summer:** approximately \$56K

Fall recorded the highest revenue, while Summer recorded the lowest among the four seasons. The relatively close values indicate that revenue was distributed fairly consistently across seasons.

Payment Method

Revenue varied across payment methods. **Credit Card** generated approximately \$87.38K (37.5%), followed by **UPI at \$62.72K (26.9%)** and **PayPal at \$41.65K (17.9%)**.

Debit Card contributed approximately \$24.51K (10.5%), while **Cash on Delivery** contributed approximately \$16.82K (7.2%).

Credit Card was therefore the largest revenue-generating payment method in the analysis.

Shipping Type

Standard Shipping generated the highest revenue at approximately \$99.34K (42.7%).

This was followed by **Free Shipping at \$78.81K (33.8%)** and **Express Shipping at \$54.93K (23.6%)**.

Standard Shipping therefore accounted for the largest share of revenue among the shipping types analyzed.

Discount Performance

Purchases where a **discount was applied** generated approximately **\$126.94K (54.4%)** of revenue, while purchases with **no discount** generated approximately **\$106.14K (45.6%)**.

The dashboard therefore shows a higher revenue contribution from purchases where discounts were applied.

Location Performance

Among the Top 10 locations by revenue, **New York** generated the highest revenue at approximately **\$13.59K**, followed by **Los Angeles at \$12.52K** and **Chicago at \$11.64K**.

Other high-revenue locations included Houston, Phoenix, San Francisco, Seattle, Miami, Dallas, and Atlanta.

Product Size

Revenue was also analyzed across product sizes. **Medium (M)** generated the highest revenue at approximately **\$105.17K (45.1%)**, followed by **Large (L) at approximately \$61.67K (26.5%)**.

This indicates that M and L sizes contributed a larger share of revenue compared with the other sizes in the dataset.

Product Color

The analysis also compared revenue across product colors. **Green** was among the highest-revenue colors at approximately **\$11K**, while several other colors, including Teal, Yellow, Olive, and Cyan, generated approximately **\$10K each**.

This provides additional information that can be considered when evaluating product assortment and color preferences.

Overall Insight

Overall, the analysis shows that revenue is concentrated across certain product categories, customer groups, products, locations, and purchasing methods. **Clothing was the strongest product category, the 40–59 age group was the highest-revenue age group, and the Occasional and Regular customer segments contributed the largest segment-level revenues.** Credit Card was the leading payment method, while Standard Shipping accounted for the largest share of shipping-related revenue.

These findings provide a useful basis for evaluating product performance, customer targeting, promotional strategies, and sales planning.

16. BUSINESS RECOMMENDATIONS

Based on the findings from the customer shopping and revenue analysis, the following recommendations can be considered.

16.1 Focus on High-Performing Categories

Clothing and Accessories generated the highest category-level revenue. These categories can receive greater attention in inventory planning, marketing, and promotional activities.

16.2 Promote High-Performing Products

Products generating higher revenue can be given greater visibility through product placement, promotions, and marketing campaigns.

16.3 Use Customer Segmentation

The differences in revenue across customer segments can be used to design more targeted marketing and customer retention strategies.

16.4 Use Demographic Insights

The differences in revenue across age groups and genders can support more focused customer targeting and marketing campaigns.

16.5 Encourage Repeat Purchases

Customers with lower purchasing frequency can be targeted through personalized offers, loyalty campaigns, and other strategies designed to encourage repeat purchases.

16.6 Review Discount Strategies

Discount-related revenue should be monitored regularly to understand how discounting is associated with sales performance.

16.7 Improve Seasonal Planning

Revenue was relatively consistent across seasons, but seasonal analysis can still support inventory and promotional planning.

16.8 Consider Product Attributes

Size and color performance can be considered when planning product assortment and inventory allocation.

17. PROJECT CHALLENGES AND LIMITATIONS

17.1 No Transaction Date

The dataset does not contain a transaction date or purchase date.

Because of this, the project could not perform:

- Monthly revenue trends
- Weekly revenue trends
- Year-over-year growth
- Month-on-month growth

Therefore, the project focuses on categorical, demographic, product, and purchasing behavior analysis.

17.2 Single Dataset

The project uses a single primary dataset.

A more complex business intelligence environment could include separate:

- Customer tables
- Product tables
- Date tables
- Transaction tables
- Location tables

However, this was not necessary for the current dataset.

17.3 Limited Historical Information

Without transaction dates, the analysis cannot determine how customer behavior changes over time.

17.4 Dataset Scope

The conclusions are based on the available 3,900-record dataset and therefore should be interpreted within the context of this dataset.

18. CONCLUSION

The Customer Shopping Behavior Analysis project demonstrates an end-to-end Data Analytics workflow using **Excel, Python, SQL, and Power BI**.

Excel was used to initially inspect the dataset, Python supported data preparation and exploratory analysis, SQL was used to query and analyze the prepared data, and Power BI was used to create the final interactive dashboard.

The project analyzed customer demographics, purchasing frequency, customer segments, products, categories, revenue, seasons, payment methods, shipping types, discounts, locations, sizes, and colors.

Six DAX measures were created to support the Power BI analysis:

- Average Age
- Average Purchase
- Average Rating
- Total Customers
- Total Purchases
- Total Revenue

The final Power BI solution contains **four dashboard pages**, combining detailed analysis with a final executive-level summary.

The project demonstrates how raw customer data can be transformed into structured analytical information and presented through an interactive Business Intelligence dashboard.

19. SKILLS DEMONSTRATED

Data Analytics

- Data inspection

- Data cleaning
- Data validation
- Exploratory Data Analysis
- Business analysis
- Data interpretation

Excel

- Dataset inspection
- Initial validation
- Data understanding

Python

- Data loading
- Data inspection
- Data cleaning
- Exploratory Data Analysis
- Numerical analysis
- Categorical analysis

SQL

- Data querying
- Filtering
- Aggregation
- Grouping
- Business-oriented analysis
- Revenue analysis
- Customer analysis
- Product analysis

Power BI

- Data preparation
- Power Query
- Data modeling
- DAX
- KPI creation
- Data visualization

- Interactive dashboards
- Slicers
- Dashboard formatting

Analytical Skills

- Customer behavior analysis
- Revenue analysis
- Product analysis
- Customer segmentation
- Demographic analysis
- Purchasing pattern analysis
- Business insight generation
- Data-driven recommendations

20. FUTURE ENHANCEMENTS

20.1 Time-Series Analysis

If transaction dates are added to the dataset, future versions could include:

- Monthly revenue
- Weekly sales
- Revenue trends
- Year-over-year growth
- Month-on-month growth

20.2 Customer Lifetime Value

Customer Lifetime Value analysis could be introduced to identify customers who provide greater long-term business value.

20.3 Predictive Analytics

Machine learning could be used in a future version to predict:

- Customer purchase behavior
- Customer churn
- Future purchase amount
- High-value customers

20.4 Advanced Customer Segmentation

Clustering techniques could be used to create more advanced customer segments based on:

- Spending

- Purchase frequency
- Previous purchases
- Customer engagement

20.5 Automated Refresh

The dashboard could be connected to a regularly updated data source to support automated data refresh.

20.6 Recommendation System

A future version could recommend products based on customer purchasing behavior.

21. PROJECT SUMMARY

Customer Shopping Behavior Analysis

Project Type:

Data Analytics / Business Intelligence

Dataset:

Customer Shopping Behavior Dataset

Records:

3,900

Attributes:

18

Tools:

Excel | Python | SQL | Power BI | DAX

Workflow:

Excel → Python → SQL → Power BI

Dashboard Pages:

4

DAX Measures:

6

Main Analysis Areas:

- Revenue
- Customer demographics
- Customer segmentation
- Purchasing frequency
- Subscription behavior
- Product performance
- Location performance

- Payment methods
- Shipping types
- Discounts
- Seasons
- Sizes
- Colors
- Products

Key Dashboard KPIs:

- Total Revenue: ~**\$233.08K**
- Total Purchases: ~**3.9K**
- Average Purchase: ~**\$59.76**
- Average Rating: ~**3.75**
- Total Customers: calculated using distinct Customer IDs

Project Outcome

The project transformed a raw customer shopping dataset into an end-to-end analytical solution using Excel, Python, SQL, and Power BI.

The final solution provides a structured workflow from data inspection and preparation to exploratory analysis, SQL querying, KPI development, interactive visualization, and business reporting.

The resulting Power BI dashboard allows users to explore customer behavior, revenue performance, product performance, and purchasing patterns through interactive KPIs, charts, and slicers.